NEWTON SCHOOL OF TECHNOLOGY

Data Visualization & Analytics — Capstone 2

Telecom Customer Analytics

Churn Prediction, Engagement Analysis & Market Intelligence

Sector: Telecommunications | Team ID: SectionC_G5

Member	Role
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1. Executive Summary

This report presents a full-spectrum analytics study of the Indian telecom market, covering 243,553 customer records across four operators — Airtel, BSNL, Reliance Jio, and Vodafone — spanning 28 states from 2020 to 2023. The central business problem is a persistent 20.05% customer churn rate, translating to 48,827 lost customers and significant revenue leakage.

Problem

The telecom market shows near-uniform churn across all operators (~20%), suggesting a structural market problem rather than operator-specific service failures. The objective is to identify which customer segments are churning, understand the behavioural and demographic patterns, and recommend data-driven retention interventions.

Approach

We built a Python ETL pipeline to clean and transform the raw dataset, performed EDA and statistical correlation analysis, computed a KPI framework, and delivered four interactive Tableau dashboards covering market overview, churn analysis, engagement distribution, and customer segmentation.

Key Insights

- Churn rate is 20.05% — consistent across all 4 operators (19.86%–20.37%)
- Airtel leads absolute churn at 12,404 customers; average age of churned customer is 46.1 years
- Male customers account for 29,022 churned records vs. 19,805 female, though female churn rate is marginally higher (20.30% vs. 19.88%)
- Jharkhand (21.12%), Karnataka (20.71%), and Mizoram (20.65%) are the highest-churn states
- Engagement score shows near-zero correlation with churn — current metrics are insufficient retention predictors

Key Recommendations

- Deploy demographic-targeted retention campaigns for Elderly and Male segments
- Prioritise geographic interventions in Jharkhand, Karnataka, and Mizoram
- Redesign the engagement KPI framework to incorporate RFM (Recency, Frequency, Monetary) signals
- Introduce loyalty pricing at the 2-year tenure mark to address the uniform churn cliff
- Conduct competitive pricing audit across all four operators to address plan-switching behaviour

2. Sector and Business Context

Sector Overview

India is the second-largest telecommunications market in the world, with over 1.2 billion mobile subscribers as of 2024. The market is dominated by four operators — Airtel, Reliance Jio, Vodafone-Idea, and BSNL — competing fiercely on price, data packages, and coverage. The ARPU (Average Revenue Per User) has been under sustained pressure since the entry of Reliance Jio in 2016, which triggered a price war that permanently compressed margins.

Key sector trends include rapid shift to data consumption over voice, growing rural penetration, and the rollout of 5G infrastructure. Customer acquisition costs are high, making retention the more cost-efficient lever for revenue growth.

Decision-Maker / Stakeholder

Primary stakeholders for this analysis are Chief Marketing Officers and VP-level Customer Retention heads at each operator. Secondary stakeholders include regional heads responsible for state-level performance and data science teams building churn prediction models.

Why This Problem Matters

At a 20.05% churn rate across 243,553 customers, approximately 48,827 customers are churning per analysis cycle. Assuming an average annual income proxy of INR 85,021 per customer and even a modest monthly plan ARPU of INR 500, each churned customer represents INR 6,000 in annual revenue loss. Across 48,827 customers, that is approximately INR 29.3 crore in annual revenue at risk. A 10% reduction in churn would recover approximately INR 2.93 crore per year.

3. Problem Statement and Objectives

Formal Problem Definition

Despite equal market presence across four major operators, the Indian telecom market sustains a structural churn rate of approximately 20%. This analysis seeks to determine: (a) which demographic and geographic segments exhibit above-average churn, (b) whether usage behaviour and engagement metrics can predict churn, and (c) what targeted interventions can reduce churn by at least 10% within two quarters.

Scope

- 243,553 customer records from 2020 to 2023
- 4 telecom operators: Airtel, BSNL, Reliance Jio, Vodafone
- 28 Indian states, 96 cities
- Variables: age, gender, income, usage (calls, SMS, data), engagement score, registration year
- Out of scope: network quality data, competitor plan data, customer service interaction logs

Success Criteria

- Identify at least 3 demographic segments with statistically distinguishable churn rates
- Quantify geographic churn variation across 28 states
- Produce 3-5 actionable, evidence-backed business recommendations
- Deliver an interactive Tableau dashboard answering real business questions

4. Data Description

Source

Dataset: Indian Telecom Customer Dataset sourced from Kaggle (raw, minimally processed). The dataset represents synthetic but realistic telecom customer records for the Indian market, containing row-level customer data with demographic, behavioural, and outcome variables.

Dataset Size and Coverage

Total Records	243,553 rows
Total Columns (raw)	19 columns
Time Range	January 2020 – December 2023
Operators	Airtel, BSNL, Reliance Jio, Vodafone
States	28 Indian states
Cities	96 cities
Missing Values	Zero across all columns

Key Columns

Column	Type	Description
customer_id	Integer	Unique customer identifier (Primary Key)
Telecom Partner	Categorical	Operator: Airtel, BSNL, Reliance Jio, Vodafone
Gender	Categorical	M = Male, F = Female
Customer Age	Integer	Age in years (18–74)
Region/State	Categorical	28 Indian states
Annual Income	Integer	Annual income in INR (20k–150k)
Monthly Calls	Integer	Avg calls per month
Monthly SMS	Integer	Avg SMS per month
Data Usage (GB)	Integer	Avg data consumed per month in GB
Churn Flag	Binary	Target variable: 0 = Retained, 1 = Churned
Total Engagement Score	Float	Composite usage-based engagement index
Age Segment	Categorical	Young Adult, Middle Aged, Senior, Elderly

Data Quality Issues

The dataset was well-structured with no missing values. Minor quality observations include: (1) Data Usage (GB) shows extreme values up to 10,991 GB — retained as plausible telecom variance rather than outliers. (2) Gender is binary (M/F) — no non-binary representation. (3) The 2023 cohort has only 24,575 records versus ~73,000 in prior years, consistent with a partial-year data extract.

5. Data Cleaning and ETL Pipeline

Pipeline Overview

The ETL pipeline was built entirely in Python using Jupyter Notebooks, following the structure mandated by the DVA Capstone 2 framework. The raw dataset was committed to data/raw/ and the pipeline output was saved to data/processed/.

Major Cleaning Steps

- Step 1 — Schema validation: Verified all 19 columns, confirmed data types, checked for nulls (none found)
- Step 2 — Duplicate detection: Confirmed customer_id uniqueness across all 243,553 records
- Step 3 — Date parsing: Converted date_of_registration from string to datetime; extracted registration_year and registration_month
- Step 4 — Age segmentation: Created Age Segment column (Young Adult: 18–30, Middle Aged: 31–45, Senior: 46–60, Elderly: 61+)
- Step 5 — Churn label encoding: Created Churn Text (readable Yes/No) from binary Churn Flag
- Step 6 — Engagement score validation: Verified Total Engagement Score formula consistency
- Step 7 — Income banding: Created income band labels for Tableau visualisation
- Step 8 — Output: Saved cleaned dataset to data/processed/telecom_cleaned.csv

Assumptions Made

- Data Usage extreme values (up to 10,991 GB) were retained — interpreted as high-data enterprise or tethering users
- 2023 partial year data was not extrapolated; analysis notes it as a partial year
- Gender encoded as binary (M/F) per source data; no imputation applied

Output Dataset

Cleaned dataset: 243,553 rows × 19 columns (same row count — no records dropped). Committed to data/processed/telecom_cleaned.csv.

6. KPI Framework

KPI	Formula	Value	Business Purpose
Churn Rate	(Churned / Total) × 100	20.05%	Primary retention health metric
Retention Rate	(Retained / Total) × 100	79.95%	Measures successful retention
Total Churned	COUNT(Churn Flag = 1)	48,827	Absolute at-risk volume
Avg Annual Income	MEAN(Annual Income)	₹85,021	Customer value proxy
Avg Age (Churned)	MEAN(Age | Churned)	46.1 years	Demographic risk profiling
Operator Market Share	(Operator Count / Total) × 100	~25% each	Competitive parity check
State Churn Rate	(State Churned / State Total) × 100	19.4%–21.1%	Geographic risk zoning
Avg Engagement Score	MEAN(Total Engagement Score)	2,527	Product stickiness indicator
Gender Churn Rate	(Gender Churned / Gender Total) × 100	F: 20.30%, M: 19.88%	Segment risk delta

7. Exploratory Data Analysis

Market Overview Findings

The four operators share the market almost perfectly equally: Reliance Jio (25.10%), Airtel (25.01%), Vodafone (24.96%), and BSNL (24.93%). This near-perfect market parity is unusual and suggests the dataset reflects a balanced market simulation.

Customer distribution across states is broadly even, with the top states being Uttarakhand (8,856), Karnataka (8,845), Maharashtra (8,820), and Madhya Pradesh (8,817). The top 10 cities by customer count are all from northeastern India — Agartala leads at 8,660, followed by Imphal (8,651), Aizawl (4,407), and Shillong (4,390).

Gender Distribution

Male customers constitute 59.9% of the base (145,977) versus 40.1% female (97,576). This gender skew is consistent with typical Indian telecom subscriber demographics, where mobile penetration has historically been higher among males.

Age Segment Analysis

Elderly customers (64,708) form the largest segment, followed by Seniors (64,006), Middle Aged (63,880), and Young Adults (50,959). The lower representation of Young Adults may reflect later average age of first telecom subscription in some regions, or dataset sampling choices.

Registration Trends

Registrations were roughly equal across 2020 (~73,126), 2021 (~72,926), and 2022 (~72,926). The 2023 figure (24,575) confirms this is a partial-year extract. Registration by month shows consistent inflows across January through December, with no strong seasonal spike — suggesting organic, steady-state growth.

Churn Distribution

Of 243,553 total customers, 48,827 have churned (20.05%) and 194,726 are retained. The churn distribution by operator is strikingly uniform: Airtel 12,404, Reliance Jio 12,234, Vodafone 12,129, BSNL 12,060. This uniformity is the single most important finding of the EDA — it tells us that no operator has a meaningful churn advantage.

Top 10 Cities by Customer Volume

Rank	City	Customer Count
1	Agartala	8,660
2	Imphal	8,651
3	Aizawl	4,407
4	Shillong	4,390
5	Namchi	4,373
6	Gangtok	4,316
7	Dimapur	4,313

Rank	City	Customer Count
8	Tura	4,294
9	Lunglei	4,282
10	Churachandpur	~4,200

8. Statistical Analysis

Correlation Analysis

Pearson correlation coefficients were computed between all numerical features and the Churn Flag to assess predictive power:

Feature	Correlation with Churn	Interpretation
Annual Income	-0.0033	Negligible negative — income is not a churn driver
Monthly SMS	-0.0031	Negligible — SMS frequency does not predict churn
Num Dependents	-0.0025	Near-zero — household size irrelevant to churn
Data Usage (GB)	+0.0008	Near-zero — data volume does not predict churn
Engagement Score	+0.0008	Near-zero — composite score has no predictive power
Customer Age	+0.0008	Negligible — age is not a significant churn driver
Monthly Calls	+0.0018	Negligible — call frequency has minimal churn link

Conclusion: No individual feature in this dataset shows meaningful linear correlation with churn. This is a critical finding — churn is uniformly distributed across income, age, usage, and engagement levels. This points to external drivers (pricing, competition, network coverage) rather than measurable customer behaviour as the primary churn cause.

Churn Rate by Segment

Segment	Category	Churn Rate
Operator	Airtel	20.37%
Operator	Reliance Jio	20.02%
Operator	Vodafone	19.95%
Operator	BSNL	19.86%
Gender	Female	20.30%
Gender	Male	19.88%
Age Segment	Middle Aged	20.35%
Age Segment	Elderly	20.12%
Age Segment	Senior	19.90%
Age Segment	Young Adult	19.76%

Geographic Churn Analysis

State-level churn rates range from approximately 19.4% to 21.1%. The top churning states are:

- Jharkhand: 21.12%
- Karnataka: 20.71%
- Mizoram: 20.65%
- Uttarakhand: 20.42%
- Himachal Pradesh: 20.41%

While these differences are modest in absolute terms (1–2 percentage points above the mean), at the state scale they represent hundreds of additional churned customers that could be targeted with geographic interventions.

Engagement Score vs. Churn

The average engagement score for churned customers is 2,529.21, virtually identical to retained customers at 2,526.27. This null result is statistically significant — it confirms that the current engagement score metric is not a useful retention predictor. The metric needs redesign to incorporate recency and plan-value signals.

9. Dashboard Walkthrough

All four dashboards are published on Tableau Public. Link:
https://public.tableau.com/views/telecom_analysis/Dashboard1

Dashboard 1 — Telecom Market Overview

Purpose: Executive-level snapshot of market health. KPI tiles display Total Customers (243,553), Churn Rate (20.05%), Avg Annual Income (₹85,021), States Covered (28), and Telecom Partners (4). Supporting visualisations include a pie chart of market share by operator, a gender bar chart, a top-10 city bar chart, a stacked bar of customers by age segment, and a choropleth map of customer distribution by Indian state.

Dashboard 2 — Customer Churn Analysis

Purpose: Operational deep-dive into churn patterns. KPI tiles show Total Churned (48,827), Avg Age of Churned (46.1), Highest Churning Operator (Airtel, 12,404), and Total Retained (194,726). Charts cover churned vs. retained comparison, churn by operator, churn by gender, annual churn trend, and a state-level churn rate choropleth map.

Dashboard 3 — Engagement Distribution

Purpose: Understand usage and engagement behaviour. KPI tiles show Avg Data Usage (5,010 GB), Avg Monthly Calls (49.23), Avg Monthly SMS (24.09), Max Data Usage (10,991 GB), and Avg Engagement Score (2,527). Charts include usage by partner, engagement by churn status, data usage histogram, calls by age segment box plot, usage by income band, and engagement score distribution.

Dashboard 4 — Customer Segmentation

Purpose: Demographic profiling. KPI tiles show Avg Age (46.08), Avg Income (₹85,021), Average Dependents (1.997), and Cities (96). Charts cover age band distribution, gender by partner, registration trend by month, age segment by partner, dependents distribution by churn status, and income by age band.

Filters and Interactivity

All four dashboards support the following interactive filters: Registration Year slider (2020–2023), Telecom Partner dropdown, Region/State dropdown, and Gender dropdown. Filters cross-apply across charts within each dashboard, enabling drill-down analysis for any operator-state-gender combination.

10. Key Insights

The following 10 decision-oriented insights are derived from analysis and are expressed in business language:

- **Uniform churn is a structural market problem — not an operator problem.**

All four operators churn at ~20%. No operator has built a meaningful retention advantage. Industry-wide interventions (e.g. regulatory churn reporting, shared loyalty frameworks) may be more effective than operator-specific campaigns.

- **Airtel leads absolute churn volume — a function of acquisition scale, not service failure.**

Airtel's 12,404 churned customers reflects its marginally larger customer base, not a higher rate. Churn reduction at Airtel should focus on volume management, not rate correction.

- **The average churned customer is 46.1 years old — the Senior/Middle-Aged demographic churns most.**

Retention programs should prioritise age segments that are large in volume and show above-average churn rates: Middle Aged (20.35%) and Elderly (20.12%).

- **Male customers contribute 59% of churned volume — but female churn rate is higher.**

Male customers account for 29,022 churn events, but female customers churn at 20.30% vs. male 19.88%. Both segments need attention with different messaging strategies.

- **Jharkhand, Karnataka, and Mizoram are the geographic churn hotspots.**

These states exceed the national average by 0.6–1.1 percentage points. Local network quality audits, targeted pricing interventions, or channel partnership reviews should be prioritised here.

- **Income is not a churn predictor — high and low income customers churn equally.**

Churned customers earn ₹84,772 vs. retained ₹85,084. Churn is not driven by affordability. This rules out price subsidies for high-income customers as a retention lever.

- **Current engagement score is a poor churn predictor.**

Churned customers score 2,529 vs. retained 2,526 — a difference of 3 points. The metric needs a redesign incorporating behavioural signals like plan upgrades, support ticket frequency, and payment delays.

- **Northeast India represents the densest customer cluster.**

All top 10 cities by customer count are in northeastern states. This is a strategic concentration — service disruptions or competitive entry in this region would have outsized churn impact.

- **2023 churn data signals a potential improvement — but is inconclusive due to partial year.**

Only 4,917 churn events were recorded in 2023 vs. 14,644 in 2022. This cannot be interpreted as genuine improvement without a full-year 2023 dataset.

- **Market share parity (~25% each) indicates commoditised competition.**

When all four operators hold equal share and equal churn, the market is in equilibrium — competition is happening on plan features and price rather than customer experience or loyalty. Differentiation through data-driven personalisation is an open opportunity.

11. Business Recommendations

Recommendation 1: Deploy Demographic-Targeted Retention Campaigns

Target Segments: Male customers (29,022 churned), Middle Aged (20.35% churn), Elderly (20.12% churn).

Action: Design tailored loyalty programs — family bundle offers for male earners with dependents, senior-friendly plans (simplified UI, lower data minimums, priority customer service) for the 61+ segment.

Expected Impact: A 5% reduction in Male and Elderly churn translates to approximately 2,050 retained customers per cycle. At ₹6,000 ARPU, this recovers ~₹1.2 crore annually.

Recommendation 2: Geographic Interventions in High-Churn States

Target States: Jharkhand (21.12%), Karnataka (20.71%), Mizoram (20.65%).

Action: Commission network quality audits in these states. Offer 3-month plan discounts or data bonuses to at-risk customers in these regions. Partner with local retailers for on-ground engagement.

Expected Impact: Reducing Jharkhand churn from 21.1% to 20.0% would retain approximately 100+ additional customers per state. Across 3 states, this is 300+ recoveries.

Recommendation 3: Redesign the Engagement KPI Framework

Problem: The current Total Engagement Score (based on calls, SMS, data) shows zero predictive power for churn.

Action: Rebuild the engagement metric using an RFM model — Recency (days since last high-usage event), Frequency (plan renewal pattern), Monetary (ARPU trend). Add behavioural signals: customer support contacts, app login frequency, and payment delays.

Expected Impact: A predictive engagement score that identifies at-risk customers 30–60 days before churn enables proactive outreach, potentially reducing churn rate by 2–4 percentage points.

Recommendation 4: Introduce Tenure-Based Loyalty Pricing

Observation: Churn is consistent year-over-year (14,558 in 2020, 14,708 in 2021, 14,644 in 2022), suggesting customers hit a natural tenure threshold where switching feels acceptable.

Action: Introduce a 2-year loyalty reward — automatic plan upgrade, bonus data, or cashback — triggered at the 24-month mark. Operators that have implemented similar programs (e.g., Airtel Thanks) report measurably improved retention.

Expected Impact: Even a 3% reduction in churn across the 2-year cohort represents ~700 retained customers per cycle.

Recommendation 5: Competitive Pricing Audit Across All Operators

Observation: Cross-operator churn uniformity strongly suggests customers are switching for price or plan features rather than service quality failures.

Action: Commission a quarterly competitive intelligence review covering plan pricing, data allowances, roaming rates, and bundled services (OTT, payments) across all four operators. Use this to inform dynamic pricing decisions.

Expected Impact: Closing a pricing gap of even ₹50/month (vs. a competitor plan) reduces the economic incentive to churn for price-sensitive customers, estimated to represent 30–40% of churned volume.

12. Limitations and Next Steps

Data Limitations

- Synthetic dataset — findings directional; real-world telecom data would include network quality, support ticket history, plan upgrade/downgrade events, and payment behaviour, all of which are stronger churn predictors
- 2023 partial year data makes year-on-year trend analysis unreliable
- No competitor data — cannot assess what operators customers are switching to
- Gender encoded as binary; non-binary representation absent
- Pincode data not used in analysis — could enable hyper-local geographic analysis

Methodological Limitations

- Linear correlation analysis only — non-linear relationships between features and churn were not modelled
- No predictive model built (logistic regression, decision tree, or random forest) — current analysis is descriptive, not predictive
- Engagement score formula assumed as given — not independently validated

Suggested Future Work

- Build a supervised churn prediction model (XGBoost or Logistic Regression) using available features; expected AUC > 0.65 even on this dataset
- Conduct survival analysis (Kaplan-Meier) to model time-to-churn by cohort
- Enrich dataset with network quality metrics, customer service contacts, and payment history
- Segment customers using K-Means clustering on usage behaviour and build segment-specific churn models
- Deploy a real-time churn risk scoring API that flags customers crossing an engagement threshold

13. Contribution Matrix

The following table summarises each team member's contributions. GitHub commit history and PR records are the authoritative evidence for this matrix.

Member	Role	Key Contributions	Notebooks / Deliverables
Samiksha Jangid	Project Lead / Viz Lead	Repo setup, timeline management, all 4 Tableau dashboards, GitHub admin, final submission	tableau/, README.md
Gauri Jindal	Analysis Lead	Statistical analysis, churn correlation study, segment analysis, dashboard design input	04_statistical_analysis.ipynb
Abhishek Rana	ETL Lead	Python ETL pipeline, raw data commit, cleaning notebook, ETL documentation	01_extraction.ipynb, 02_cleaning.ipynb, etl_pipeline.py
Rudraksh	Data Lead	Dataset sourcing, initial EDA, data dictionary, Gate 1 proposal	03_eda.ipynb, docs/data_dictionary.md
Ritik	Strategy Lead	Problem statement, KPI framework design, business recommendations, impact estimation	reports/ (strategy sections)
Aryan	PPT & Quality Lead	Final report PDF, presentation deck, contribution matrix, quality review	reports/project_report.pdf, presentation.pdf

— End of Report —

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